

CAPE NEWENHAM, AK, USA

WMO: 703050

Lat: 58.650N

Lon: 162.067W

Elev: 541

StdP: 14.41

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25623

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -5.4	(d) -15.0	(e) 2.5	(f) -8.3	(g) -11.2	(h) 3.1	(i) -4.4	(j) 44.1	(k) 21.9	(l) 40.7	(m) 22.9	(n) 12.1	(o) 10	(p) 0.913

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.4	(c) 62.4	(d) 55.5	(e) 59.5	(f) 54.4	(g) 57.4	(h) 53.3	(i) 57.1	(j) 60.2	(k) 55.5	(l) 58.2	(m) 54.4	(n) 56.8	(o) 7.8	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.4	66.6	57.9	54.3	64.0	56.9	53.1	61.1	55.8	24.6	60.2	23.7	58.4	23.0	57.1	64.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
39.8	35.1	31.5	-9.7	66.5	4.7	4.7	-13.1	69.9	-15.8	72.7	-18.5	75.3	-21.9	78.8		
			-9.6	59.2	3.8	2.5	-12.4	61.0	-14.6	62.5	-16.7	63.9	-19.5	65.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	20.2	23.6	21.9	30.6	39.5	46.9	51.4	52.1	47.8	38.3	30.0	23.5
	DBStd	14.51	14.08	12.12	11.50	7.97	6.11	4.81	4.17	3.53	4.19	6.73	9.30	12.31
	HDD50	5481	924	739	870	583	330	116	32	17	88	362	599	821
	HDD65	10750	1389	1159	1335	1033	792	542	421	401	517	826	1049	1286
	CDD50	207	0	0	0	0	4	25	75	81	21	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	13.6	13.4	16.2	14.8	13.7	11.4	9.3	10.2	12.7	13.9	15.2	16.9	16.3
Precipitation	PrecAvg	32.7	2.6	2.4	2.0	1.7	1.9	2.0	2.8	3.2	4.1	3.7	3.5	2.9
	PrecMax	48.0	6.0	6.4	5.6	4.0	3.6	3.5	4.5	5.1	7.8	7.2	6.9	5.0
	PrecMin	25.0	0.7	0.3	0.4	0.4	0.5	0.3	1.2	0.8	0.5	0.6	0.5	0.6
	PrecStd	5.3	1.1	1.3	1.3	0.8	0.8	0.7	0.9	1.1	1.6	1.5	1.7	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	41.2	41.4	48.2	59.5	65.2	66.8	64.3	59.9	53.1	48.2	44.1
		MCWB	38.9	38.4	37.8	40.8	48.8	55.7	58.8	57.6	55.0	49.4	45.4	41.5
	2%	DB	40.3	39.3	38.9	44.5	54.2	60.3	62.9	61.1	56.4	50.6	45.3	41.0
		MCWB	37.5	36.8	35.8	39.4	46.5	52.6	56.3	56.2	52.6	47.7	42.7	38.9
	5%	DB	38.3	37.8	37.1	41.7	50.4	56.9	59.4	58.8	54.6	48.5	43.1	39.0
		MCWB	36.0	35.4	34.4	37.8	44.7	50.6	54.6	55.3	51.3	46.0	40.9	36.8
	10%	DB	36.5	36.6	35.5	39.5	47.8	54.1	57.2	57.0	53.1	46.6	41.4	37.2
		MCWB	34.6	34.4	33.3	36.5	43.4	49.3	53.4	54.2	50.2	44.2	39.5	35.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.9	39.3	38.9	42.2	50.1	57.0	60.5	59.2	56.1	50.2	45.9	42.3
		MCDB	42.5	40.7	40.5	46.2	57.0	62.2	65.7	62.3	58.1	51.9	47.5	43.6
	2%	WB	37.9	37.5	36.4	39.9	47.3	54.0	57.1	57.2	53.4	48.1	43.3	39.3
		MCDB	39.8	38.8	38.2	43.5	52.7	59.1	60.8	59.6	55.5	50.3	45.1	40.8
	5%	WB	36.4	36.0	34.8	38.2	45.5	51.7	55.5	55.8	51.9	46.3	41.5	37.2
		MCDB	38.1	37.4	36.6	41.2	49.9	55.6	58.6	57.9	54.0	48.3	43.1	38.7
	10%	WB	34.7	34.7	33.4	36.8	43.7	49.8	54.2	54.7	50.6	44.8	39.7	35.5
		MCDB	36.3	36.2	35.2	39.3	47.5	53.2	56.7	56.7	52.6	46.6	41.5	37.0
Mean Daily Temperature Range	5% DB	MDBR	8.7	8.8	8.5	8.0	7.8	8.2	6.5	5.4	5.0	5.7	7.2	8.2
		MCDBR	7.4	6.8	7.9	9.3	12.1	12.2	10.1	8.2	6.2	5.3	6.3	7.5
		MCWBR	5.8	5.5	6.0	6.2	7.6	7.1	6.1	5.2	4.9	5.1	5.8	6.4
	5% WB	MCDBR	7.0	6.9	7.5	8.7	11.3	11.4	8.6	7.3	5.6	5.1	6.1	6.7
		MCWBR	6.0	6.0	6.1	6.0	7.4	7.4	5.8	5.1	4.9	5.1	6.1	6.1
Clear-Sky Solar Irradiance	taub		0.260	0.305	0.304	0.360	0.382	0.365	0.378	0.373	0.344	0.313	0.293	0.260
	taud		2.158	2.173	2.237	2.182	2.280	2.430	2.429	2.444	2.473	2.400	2.139	2.048
	Ebn at Noon		179	224	266	266	269	276	269	262	252	224	157	128
	Edh at Noon		15	24	30	39	38	33	32	30	25	20	15	12
All-Sky Solar Radiation	RadAvg		130	335	839	1290	1611	1648	1378	1101	775	404	150	70
	RadStd		26	64	143	142	138	122	147	124	95	38	19	11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page